

Ques. HOW DOES BDMA TECHNOLOGY WORK IN 5G NETWORK?

Prof. Ashish Mulajkar

Ans. Beam-division multiple-access (BDMA) scheme has been proposed to increase wireless communication system capacity and handle a large number of users in 5G systems.

The number of mobile users is swelling day by day and is expected to grow ten times by the year 2020. Currently, techniques like time-division multiple-access (TDMA) and frequency-division multiple-access (FDMA) are adopted in communication systems. But the frequency and time division among multiple users face problems due to the increasing number of mobile users.

Korean engineers have suggested BDMA as a radio interface for 5G systems as it is independent of frequency and time resources. This technique not only solves the problem of bandwidth capacity and efficiency but also eliminates signal deterioration at the cell edge.

In BDMA technology, the base station allocates separate antenna beam to each mobile phone or mobile station. This could be done after evaluating position of the mobile stations as well as their moving speeds, then deciding widths and directions of beams for mobile stations with respect to the base station. All the mobile stations communicate with base station when these are in its line-of-sight (LoS) and all the beams are distinct for each individual station. So simultaneous data transmission is possible to different mobile stations at different angles.

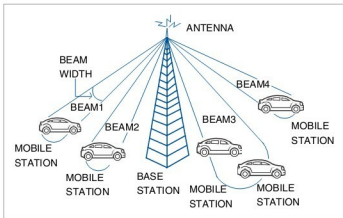


Fig. 1: Beam-division multiple-access



Fig. 2: USB Redirector

To support more number of users, the beam should be divided. Mobile stations located at similar angles can share one beam to communicate with the base station using TDMA.

Q2. HOW DO I ACCESS USB STORAGE DEVICE THROUGH THE INTERNET?

A. Samiuddin

A2. With the Internet, you can not only access a USB storage device but the whole remote computer. There are many ways to access the USB drive through computer network. In Win-

dows PC you can share the drive and enable Remote Desktop connection. There are also many software available for that, including TeamViewer and USB Redirector.

USB Redirector application allows users to access any USB device on a network. Two versions are available: USB Redirector Lite Client (freeware) for client computer and USB Redirector (shareware) for server computer.

You need to install this tool on both the computers (server and client) and connect to the Internet. Then connect the USB drive to your PC on the server side, run USB Redirector and then share the USB drive (Fig. 2). Run USB Redirector Lite Client to connect the server side. Provide the server IP address and host name in the client computer.

If connection is successful, you will see the list of shared USB devices. Select the USB device you need to connect to and then press Connect USB

Device to access the files.

Another method to access USB storage files is a router with USB port.

Connect the USB flash drive to your home router. Some routers come with a USB port where you can plug in a USB device and then access its data.

To get this setup working, you need to enable remote sharing or setup dynamics DNS (domain name system) on your router.

Answers compiled by Sani Thea, technical editor. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editesc@efy.in) and should include name and address of the sender.